CHEMISTRY

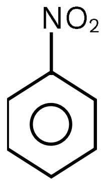
Q.1 Compare boiling point of given solutions

(I) 10^{-4} NaCl (II) 10^{-3} NaCl (III) 10^{-2} NaCl (IV) 10^{-4} urea

(1) I > II > III > IV (2) III > II > I > IV (3) II > I > III > IV (4) III > I > II > IV

Q2 Choose the incorrect order of electronegativity

(1) S < Cl < O < F (2) Se < S < O < F (3) Al < Mg < Na (4) I < Br < Cl < F

Q3  $\xrightarrow[\text{(ii) } NaNO_2/H_4 \quad \text{(iii) } CuCl/H_4 \quad \text{(iv) Na/dry ether}]{\text{(i) Sn + HCl}}$ A.

Find molecular weight of A

Q4. Number of linear compounds:

I_3^-, NO_2, O_3, OF_2, NO_2^+, $BeCl_2$, N_3^-, SO_3, SO_2, CO_2, XeF_2

Q5 The Ground state radius of Hydrogen atom is a_0. Calculate the radius of first excited state of Helium ?

Q6 Given the weight of the organic compound, 180 mg and the weight of the AgCl precipitated 143.5 gm. Calculate the estimation of Cl in percentage.
[Weight of Cl = 35.5 gm and Weight of Ag is 108 gm]

Q7 Which of the following lanthanide ion $4f^7$ electronic configuration

Eu^{+3}, Gd^{+3}, Eu^{+2}, Gd^{+2}, Sm^{+2}, Tb^{+2}

(1) Eu^{+3}, Gd^{+2} (2) Gd^{+3}, Eu^{+2} (3) Sm^{+2}, Tb^{+2} (4) Gd^{+2}, Sm^{+2}

Q8 Which of the following can be used to prepare marshall's acid ($H_2S_2O_8$)

(1) Electrolysis of Conc. Na_2SO_4 (2) Electrolysis of dilute Na_2SO_4

(3) Electrolysis of Conc. H_2SO_4 (4) Electrolysis of dilute H_2SO_4

Q9

List-I	**List-II**
(P) IE	(1) B < C < O < N
(Q) Metallic character	(2) Al < Si < P < S
(R) Electronegativity	(3) $F^- > Na^+ > Mg^{+2} > Al^{+3}$
(S) ionic radii	(4) K > Mg > Al > S

(1) P→1; Q→4; R→2; S→3 (2) P→3; Q→4; R→2; S→1

(3) P→1; Q→2; R→4; S→3 (4) P→4; Q→1; R→2; S→3

Q10 $CH_3-CH=CH-\underset{\large OH}{\underset{|}{CH}}-CH_3$

Find out the total number of stereoisomers

Q11 $CO_2 + C_{(S)} \rightleftharpoons 2CO$

Calculate Partial Pressure of $CO_{(g)}$. If $CO_{2(g)}$ at 0.5 atm is passed overheated carbon. Total pressure at equilibrium is 0.8.

Q12 Which of the following is not true?

(1) decay constant does not depends on temperature.

(2) decay constant increases with temperature.

(3) $t_{1/2} = \dfrac{\ln 2}{K}$

(4) None

Q13 Given below are two statements :

Statement (I): CH_3-O-CH_2-Cl ; 1° alkyl halide show SN^1 reaction.

Statement (II): $CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-CH_2-Cl$; show SN^2 reaction.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are correct

(4) Statement I is incorrect but Statement II is correct

Q14 Correct IUPAC naming of the following

$CH_3-\underset{COOH}{\underset{|}{CH}}-CH_2-CH_2-\underset{COOCH_3}{\underset{|}{CH}}-CH_3$

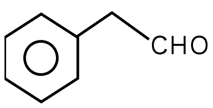
(1) 5-methoxy Carbanyl-2-methyl hexanoic acid

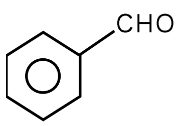
(2) 2-methoxy Carbanyl-5-methyl hexanoic acid

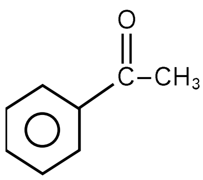
(3) 2-methyl-5-methoxy Carbanyl hexanoic acid

(4) 5-methoxy Carbanyl-1-methyl hexanoic acid

Q15 Which of the following gives positive test with Fehling's solution?

(1) 

(2) 

(3) 

(4) CH_2-OH
$|$
$C=O$
$|$
$(CH-OH)_3$
$|$
CH_2-OH

Q16 Ascorbic acid is which type of vitamin?

(1) Vitamin A (2) Vitamin B (3) Vitamin C (4) Vitamin D

Q17 $[Ni(CO)_4]$ and $[NiCl_4]^{2-}$ respectively have

(1) $[Ni(CO)_4]$ is square planar, diamagnetic while $[NiCl_4]^{2-}$ is Tetrahedral, Paramagnetic

(2) $[Ni(CO)_4]$ is tetrahedral, Paramagnetic while $[NiCl_4]^{2-}$ is Tetrahedral, diamagnetic

(3) $[Ni(CO)_4]$ is tetrahedral, diamagnetic while $[NiCl_4]^{2-}$ is Tetrahedral, Paramagnetic

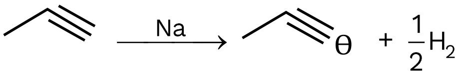
(4) $[Ni(CO)_4]$ is tetrahedral, diamagnetic while $[NiCl_4]^{2-}$ is Square planar, diamagnetic

Q18 Which of the following compound have CFSE value is zero?

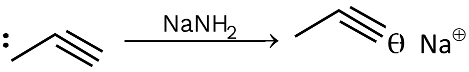
(1) $K_4[Fe(CN)_6]$ (2) $K_3[Fe(SCN)_6]$ (3) $[Fe(en)_3]Cl_3$ (4) $[Fe(NH_3)_6]Br_2$

Q19 Given below are two statements :

Statement (I):



Statement (II):



In the light of the above statements, choose the most appropriate answer from the options given below :

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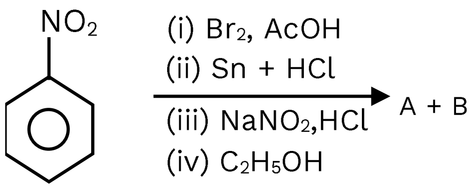
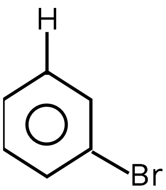
(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are correct

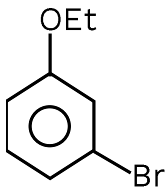
(4) Statement I is incorrect but Statement II is correct

Q20

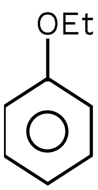


(1) A. 

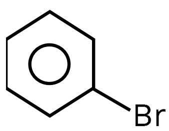
B. $CH_3-\overset{\overset{O}{||}}{C}-H$

(2) A. 

B. $CH_3-\overset{\overset{O}{||}}{C}-H$

(3) A. 

B. $CH_3-\overset{\overset{O}{||}}{C}-H$

(4) A. 

B. CH_3-COOH